

Django Workbook - Step by Step Notes

Question 1: Create Project and Task Models

Step 1: Open models.py

Step 2: Create Project model.

Fields: name, description, start_date, end_date.

Step 3: Create Task model.

Fields: title, description, status, due_date.

Step 4: Add ForeignKey if relationship is required.

Example: `added_by = models.ForeignKey(Project, on_delete=models.CASCADE)`

ForeignKey Concept

ForeignKey creates a relationship between two models.

Project = Parent Model.

Task = Child Model.

One Project can have many Tasks.

CASCADE means if Parent(Project) is deleted, related Task records are automatically deleted.

Step 5: Make Migrations

Command 1:

`python manage.py makemigrations`

Purpose: Create migration file after changing models.py.

Command 2:

`python manage.py migrate`

Purpose: Apply migration to database.

Rule: Whenever models.py changes -> Run makemigrations then migrate.

Question 2: Register Models in Django Admin

Open admin.py

Import admin module.

Import Project and Task models.

Simple Registration:

`admin.site.register(Project)`

`admin.site.register(Task)`

Meaning: Register models so they appear in Django Admin Panel.

Decorator Method

`@admin.register(Project)`

`class ProjectAdmin(admin.ModelAdmin):`

admin.ModelAdmin is Django's built-in class for admin customization.

Common attributes:

list_display -> Show columns

search_fields -> Search

list_filter -> Filter

ordering -> Sorting

list_editable -> Edit directly

list_per_page -> Pagination

readonly_fields -> Read only

Question 3: Create Superuser

Command:

python manage.py createsuperuser

Enter Username

Enter Email (optional)

Enter Password

Run Server:

python manage.py runserver

Open Browser: <http://127.0.0.1:8000/admin/>

Login and Add Project records.

Login and Add Task records.

Question 4: Django Shell

Open Shell:

python manage.py shell

Import Models:

from newapp.models import Project, Task

Fetch One Project:

project = Project.objects.get(id=1)

Meaning: Get single Project whose id is 1.

Fetch Related Tasks:

Task.objects.filter(added_by=project)

Meaning: Get all tasks related to selected Project.

get() returns one record.

filter() returns one or more records (QuerySet).

QuerySet = Collection of database records.

Useful Commands Sequence

1. Edit models.py

2. python manage.py makemigrations

3. python manage.py migrate

4. Edit admin.py
5. `python manage.py createsuperuser`
6. `python manage.py runserver`
7. Open /admin
8. Add records
9. `python manage.py shell`
10. Query database using ORM.